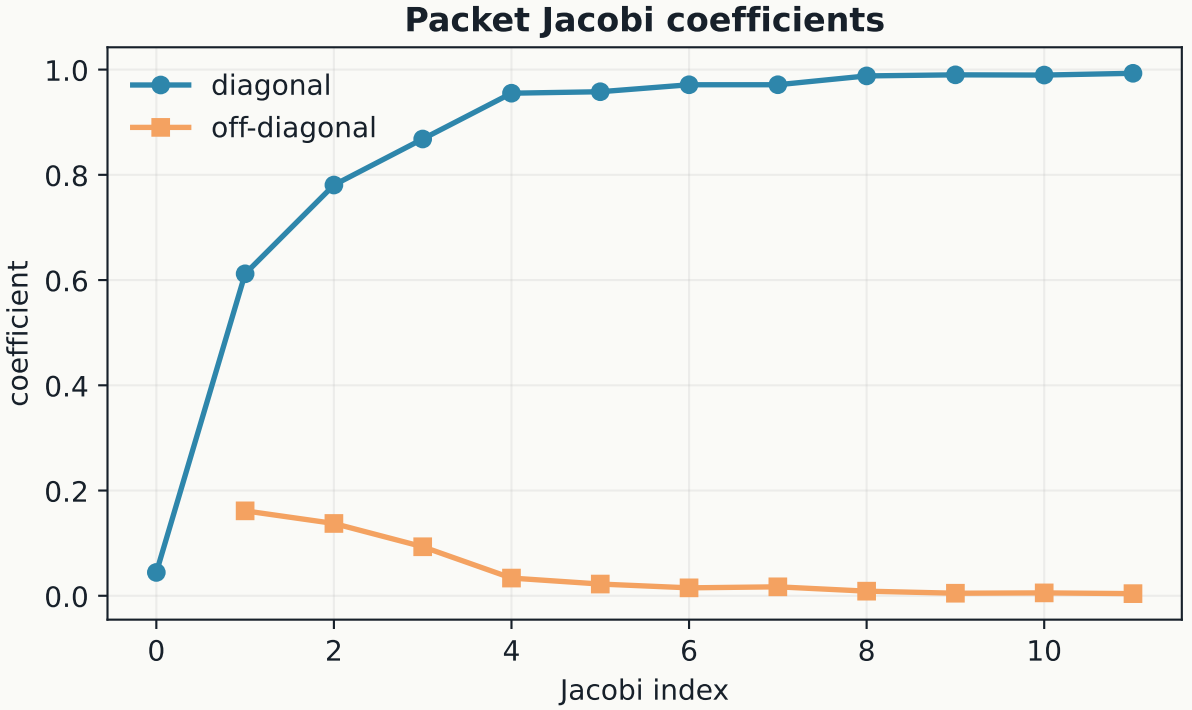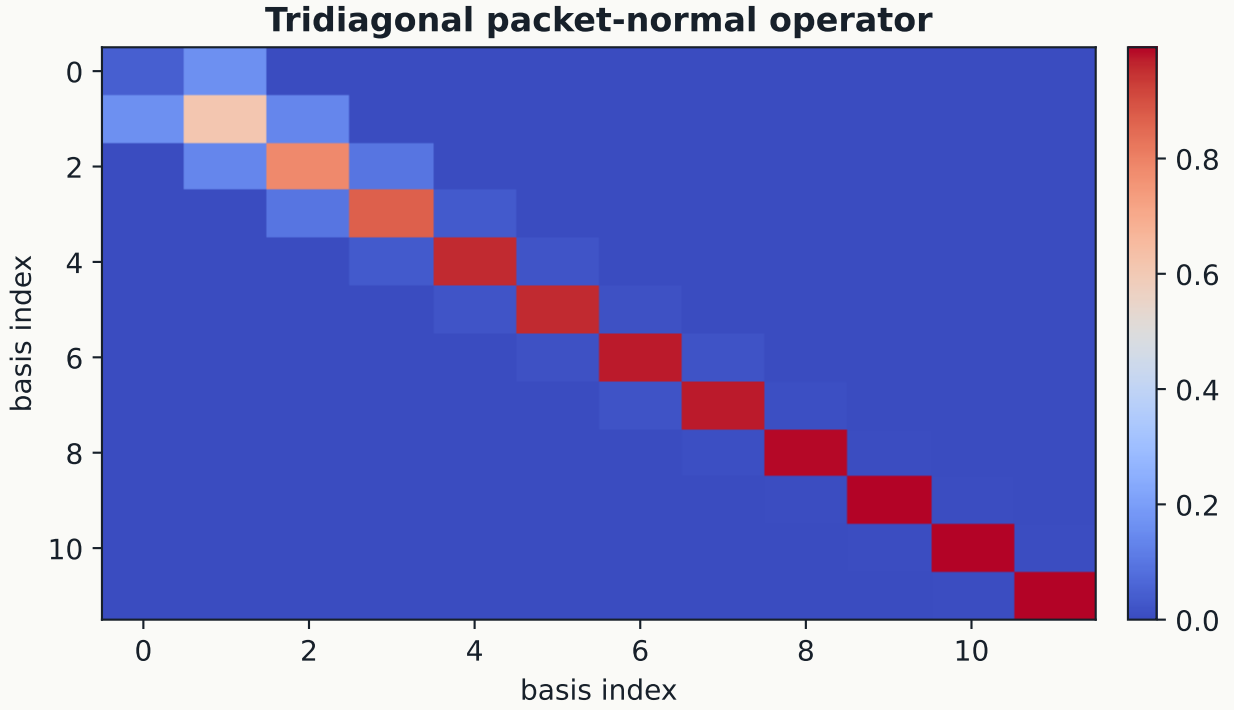Lanczos converts the diagonal number-basis contraction into one canonical Jacobi matrix